

Web System Technologies

IT0049



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Module 2

CodeIgniter Installation and Configuration



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Objectives

- To introduce CodeIgniter as an MVC framework.
- To familiarize student on CodeIgniter's features, application flow chart, concepts, design and architectural goals.
- To learn how to install CodeIgniter 4
- To be able to configure the environment settings
- To explore the directory structure



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CodeIgniter

CodeIgniter is an open-source web framework, for creating web sites with PHP and is loosely based on model–view–controller (MVC) development pattern.

CodeIgniter can be also modified to use Hierarchical Model View Controller (HMVC).

CodeIgniter at a Glance

- Is an Application Framework
- Free
- Light weight
- Fast
- Uses M-V-C
- Generates clean URLs
- Extensible
- Does Not Require a Template Engine
- Thoroughly Documented



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CodeIgniter MVC

MVC is a software approach that separates application logic from presentation. In practice, it permits your web pages to contain minimal scripting since the presentation is separate from the PHP scripting.

- The **Model** represents your data structures.
- The **View** is the information that is being presented to a user.
- The **Controller** is the *intermediary* between the Model, the View, and any other resources.



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CodeIgniter Features

- Model-View-Controller Based System
- Extremely Light Weight
- Full Featured database classes with support for several platforms.
- Query Builder Database Support
- Form and Data Validation
- Security and XSS Filtering
- Session Management



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CodeIgniter Features

- Email Sending Class.
- Image Manipulation Library
- File Uploading Class
- FTP Class
- Localization
- Pagination



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CodeIgniter Features

- Data Encryption
- Benchmarking
- Full Page Caching
- Error Logging
- Application Profiling
- Calendaring Class
- User Agent Class
- Zip Encoding Class



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CodeIgniter Features

- Template Engine Class
- Trackback Class
- XML-RPC Library
- Unit Testing Class
- Search-engine Friendly URLs
- Flexible URI Routing
- Support for Hooks and Class Extensions
- Large library of “helper” functions



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Design and Architectural Goals

CodeIgniter's goal is to obtain maximum performance, capability, and flexibility.

CodeIgniter was created with the following objectives:

- Dynamic Instantiation
- Loose Coupling
- Component Singularity



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Software Requirements

- **OS (Cross Platform)**
- **Web Browser**
- **Web Server (Apache)**
- **Text Editor**



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CodeIgniter Setup

PHP Version Requirement

PHP Version 7.3 or Higher: CodeIgniter 4 requires at least PHP 7.3. This version introduces several new features and optimizations that CodeIgniter 4 relies on, including improved performance and security enhancements.



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Installation

CodeIgniter has two supported installation methods: manual download, or using [Composer](#).

- CodeIgniter recommends the Composer installation because it keeps CodeIgniter up to date easily.
- If you would like the simple “download & go” install that CodeIgniter 3 is known for, choose the manual installation.

Composer Installation

In the folder above your project root run:

```
composer create-project codeigniter4/appstarter project-root
```

The command above will create a **project-root** folder. If you omit the “project-root” argument, the command will create an “appstarter” folder, which can be renamed as appropriate.



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Manual Installation

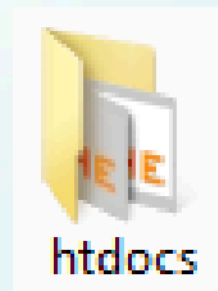
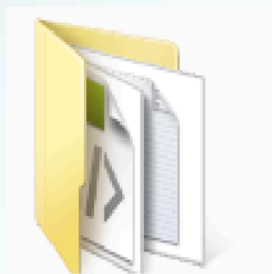
1. Visit the [CodeIgniter 4 official website](#) or the [GitHub repository](#).
2. Navigate to the “Downloads” section.
3. Download the latest stable release as a ZIP file.



Manual Installation

After downloading the ZIP file, it needs to be extracted to your web server's root directory.

1. Locate the downloaded ZIP file on your system.
2. Right-click on the ZIP file and select “Extract All...” or use a file extraction tool like WinRAR, 7-Zip, or macOS built-in utility.
3. Extract the contents to a directory within your web server's document root (e.g., htdocs for XAMPP or www for WAMP).



Initial Configurations

Configuring the environment:

CodeIgniter 4 requires some initial configuration to work correctly.

1. Rename the env file (located in the root directory) to .env. This file contains environment-specific settings.
2. Open the .env file in a text editor.
3. Set the CI_ENVIRONMENT variable to development or production, depending on your needs.

```
#-----  
# Example Environment Configuration file  
#  
# This file can be used as a starting point for your own  
# custom .env files, and contains most of the possible settings  
# available in a default install.  
#  
# By default, all of the settings are commented out. If you want  
# to override the setting, you must un-comment it by removing the '#'  
# at the beginning of the line.  
#-----  
  
#-----  
# ENVIRONMENT  
#-----  
  
CI_ENVIRONMENT = development
```



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Initial Configurations

Setting up the BASE URL:

The base URL configuration tells CodeIgniter where the application is located.

1. Open app/Config/App.php in a text editor.
2. Locate the \$baseURL variable and set it to the URL where your application will be accessed.

```
public $baseURL = 'http://localhost/your-project-folder/';
```

If you need more flexibility, the baseURL may be set within the [.env](#) file as `app.baseURL = 'http://localhost/your-project-folder/'`.

Always use a trailing slash on your base URL!

Initial Configurations

Setting up the Index Page:

If you don't want to include **index.php** in your site URIs, set **\$indexPath** to **"**. The setting will be used when the framework generates your site URIs.



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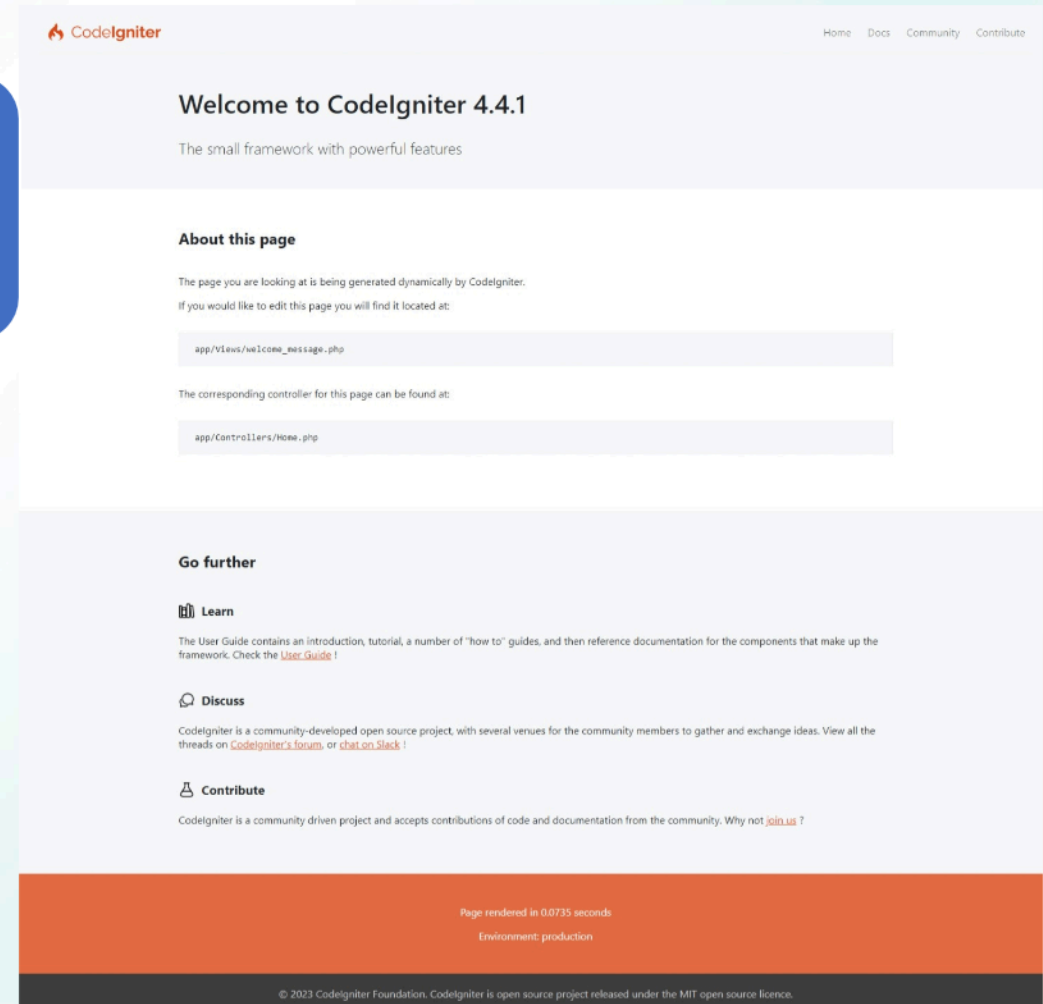
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Verifying Installation

After setting up, verify that CodeIgniter 4 is installed correctly.

1. Open a web browser and navigate to `http://localhost/your-project-folder/` (replace your-project-folder with the actual folder name).
2. You should see the default CodeIgniter welcome page if the installation is successful.



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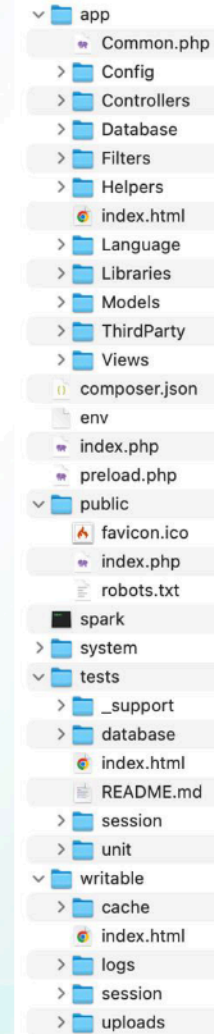


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Exploring the Directory Structure

1. Root Directory
2. Application Directory
3. System Directory
4. Public Directory

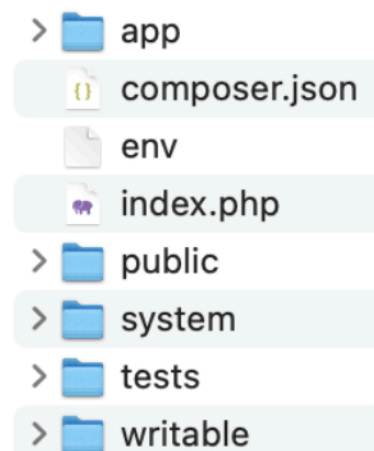


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Root Directory

The root directory contains the main files and folders of the CodeIgniter 4 application.

- `index.php`: The entry point for the application. initializes the framework and loads the appropriate components based on the request.
- `env`: Environment file (usually renamed to `.env`), which contains environment-specific settings.
- `composer.json`: Composer configuration file that manages dependencies.
- `Application (app) directory`: Contains the application's code, including controllers, models, views, and configuration files.
- `Public directory`: Where publicly accessible files are stored. It serves as the entry point for web requests.
- `System directory`: Contains the core framework files of CodeIgniter 4.
- `Tests directory`: This directory is set up to hold your test files.
- `Writable directory`: Stores cache files, logs, and any uploads user might send.



```
> app
> composer.json
> env
> index.php
> public
> system
> tests
> writable
```


Application (app) Directory

The **app** directory is where all of your application code lives. This comes with a default directory structure that works well for many applications.

The following folders make up the basic contents:

app/

Config/	Stores the configuration files
Controllers/	Controllers determine the program flow
Database/	Stores the database migrations and seeds files
Filters/	Stores filter classes that can run before and after controller
Helpers/	Helpers store collections of standalone functions
Language/	Multiple language support reads the language strings from here
Libraries/	Useful classes that don't fit in another category
Models/	Models work with the database to represent the business entities
ThirdParty/	ThirdParty libraries that can be used in application
Views/	Views make up the HTML that is displayed to the client

System Directory

This directory stores the files that make up the framework, itself.

While you have a lot of flexibility in how you use the application directory, the files in the system directory should never be modified. Instead, you should extend the classes, or create new classes, to provide the desired functionality.

All files in this directory live under the CodeIgniter namespace.



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Public Directory

The **public** folder holds the browser-accessible portion of your web application, preventing direct access to your source code.

It contains the main **.htaccess** file, **index.php**, and any application assets that you add, like CSS, javascript, or images.

This folder is meant to be the “web root” of your site, and your web server would be configured to point to it.



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Common Config Options

- **Base URL:** app/Config/App.php
- **Database Settings:** app/Config/Database.php
- **Error Handling:** app/Config/Logger.php
- **Session Management:** app/Config/Session.php



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CodeIgniter URLs

Overview

By default, URLs in CodeIgniter are designed to be search-engine and human-friendly. Rather than using the standard “query-string” approach to URLs that is synonymous with dynamic systems, CodeIgniter uses a **segment-based** approach:

https://example.com/news/article/my_article



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URL Structure

Base URL contains only the Hostname:

When you have the Base URL <https://www.example.com/> and imagine the following URL:

<https://www.example.com/blog/news/2022/10?page=2>

Term	Example	Description
Base URL	https://www.example.com/	Base URL is often denoted as baseURL .
URI path	/blog/news/2022/10	
Route path	/blog/news/2022/10	The URI path relative to the Base URL. It is also called as URI string .
Query	page=2	

URL Structure

Base URL contains sub folders:

When you have the Base URL <https://www.example.com/ci-blog/> and imagine the following URL:

<https://www.example.com/ci-blog/blog/news/2022/10?page=2>

Term	Example	Description
Base URL	https://www.example.com/ci-blog/	Base URL is often denoted as baseURL .
URI path	/ci-blog/blog/news/2022/10	
Route path	/blog/news/2022/10	The URI path relative to the Base URL. It is also called as URI string .
Query	page=2	



URI SECURITY

CodeIgniter is fairly restrictive regarding which characters it allows in your URI strings (Route path) in order to help minimize the possibility that malicious data can be passed to your application. URIs may only contain the following:

- Alpha-numeric text (latin characters only)
- Tilde: ~
- Percent sign: %
- Period: .
- Colon: :
- Underscore: _
- Dash: -
- Space: " "



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Removing the index.php file

When you use Apache Web Server, by default, the **index.php** file will be needed in your URLs:

example.com/index.php/news/article/my_article

If your server supports rewriting URLs you can easily remove this file with URL rewriting. This is handled differently by different servers, but we will show examples for the two most common web servers here.

Apache Web Server

Apache must have the *mod_rewrite* extension enabled. If it does, you can use a **.htaccess** file with some simple rules. Here is an example of such a file, using the “negative” method in which everything is redirected except the specified items:

```
RewriteEngine On
RewriteCond %{REQUEST_FILENAME} !-f
RewriteCond %{REQUEST_FILENAME} !-d
RewriteRule ^(.*)$ index.php/$1 [L]
```



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